Avi Aggarwal

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EDUCATION

Colorado School of Mines, Golden, CO — B.S. Mechanical Engineering — **December 2027**

Honors and Awards: Oredigger Leadership Community

Allen High School, Allen, TX — High School Diploma with STEM Endorsement — **May 2023**

Honors and Awards: Boeing STEM Scholar, NASA HUNCH Finalist, AP Scholar

PROJECTS

- **PID Easy-Bake Oven** (MEGN300): Designed and built an Easy-Bake oven and developed a PID controller in LabVIEW to regulate temperature, successfully baking a cookie in approximately 8 minutes.
- **Arduino Lockbox** (MEGN200): Programmed and integrated a fingerprint sensor with an Arduino microcontroller to create a biometric security system; developed circuitry, wiring, and control logic for the lock. Also modeled the overall housing for electrical components using SolidWorks.
- **Derby Car** (MEGN201): Produced detailed GD&T drawings and performed tolerance stack-up analysis; fabricated and assembled components; tested performance against design specifications and validated compliance.
- **Suitcase Redesign** (EDNS151): Applied CAD modeling and rapid prototyping to redesign a suitcase for improved usability; developed a mobile app in support of product functionality; contributed to design evaluation and iteration.
- **NASA Hunch Micro-G Dice Rolling**: Developed a magnetic dice-rolling blaster with NASA's HUNCH program, leading testing and prototyping efforts that earned Design Review Finalist recognition and selection for MIT's reduced-gravity aircraft.

RELEVANT EXPERIENCE

Substation Field Engineering Intern, Xcel Energy, Denver, CO

- Designed and implemented a Microsoft Power Apps workflow to improve application notification management and streamline internal processes.
- Developed an AI-based agent to improve the classification of substation maintenance notifications and support more efficient work prioritization.
- Supported substation field engineering activities by interpreting one-line diagrams and transformer nameplates and assisting with breaker replacement projects.
- Analyzed historical field labor and overtime hours to support five-year workforce cost forecasting and long-term budget planning.
- Used Microsoft Dataverse and SharePoint to review, update, and modernize engineering standards and internal workflows.
- Assisted in developing safety presentations and technical materials for engineering and field personnel.
- Gained practical knowledge of substation design, equipment, maintenance practices, and utility field engineering operations.

Compressor Engineering Intern, Honeywell-Sundyne, Arvada, CO

- Designed and demoed a new Compressor Test Database, improving usability and accuracy.
- Contributed to the EPR 1451 LO Vortex Study through test stand design, load calculations, pump sizing, and part sourcing.
- Built organized Inducer and Pump Case databases to streamline sizing and analysis requests.
- Created Item Masters in the part management system to support engineering workflows.
- Collaborated across teams to strengthen technical expertise in compressors, fluid mechanics, and machine design.

Computer Science Teaching Assistant, Colorado School of Mines, Golden, CO

- Works with professors to create grading rubrics for assignments
- Assists students with Python-based programming and code reviews
- Holds office hours where students can come and ask questions
- Helps write assignments for students and ensures that they target key topics

Mines Activities Council Movie Night Chair, Colorado School of Mines, Golden, CO

- Interviewed and joined the committee as 2024-2025 Movie Night Chair
- Work with a movie licensing company to get permission to show movies on campus
- Manage budget and work with a team to put on large events on campus
- Organizes vendors for events and manages quotes
- Created a new budget management system and assisted with bugs and questions

ENGINEERING AND TECHNICAL SKILLS

- **Technologies:** SolidWorks, Microsoft Office, 3D Printing, Python, C++, Arduino, LabVIEW, MATLAB, EES, Dataverse, SharePoint
- **Certifications:** OSHA 30-Hour General Industry Safety and Health, Solid Edge Associate Level, CPR
- **Professional:** Machine Shop Experience, Problem Solving, Public Speaking, Management, Leadership

ACTIVITIES

- American Society of Mechanical Engineers (ASME)
- Society of Asian Scientists and Engineers (SASE)
- South Asian Student Association at Mines (SASA)